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2-DOF Robot Arm with Variable Torque Limiters Realized by Electrostatic Film Motors

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***Abstract*— When a conventional robot arm, driven by magnetic motors and reduction gears, operates at high speed, it can produce an excessive force due to unexpected contact. Toward the development of a safe robot arm, the present study investigated a 2-degree-of-freedom robot arm driven by direct-drive electrostatic film motors. The motors operate synchronously, except when they are subjected to an excessive force. This loss of synchronicity is referred to as a "step-out" process, and in the present study it was used to produce a variable torque limiter. By setting appropriate limits, contact forces caused by unexpected collisions could be suppressed. The torque limiter was applied to a robot arm with a length of 50 cm. Experiments showed that the arm was impact-resistant; when it was struck by a hammer, the torque limiter was activated and suppressed the collision force, allowing the robot arm to successfully resume its motion. When the robot arm unexpectedly hit an obstacle while moving at 1.2 m/s, the collision force was maintained at 21 N by the torque limiter, which is sufficiently small for a high-speed collision. The experiments verified the effectiveness of the electrostatic film motor as a variable torque limiter to realize safe contact.**

I. INTRODUCTION

Collaborative robot arms have been developed to operate in a human environment [1], [2], [3]. These arms are relatively lightweight and are designed with human safety in mind [4]. However, the arms are still relatively heavy, and may cause injury if they collide with a human at high speed. Therefore to ensure safety, the operating speed (hand speed) is restricted to a low value such as 0.25 m/s [5], [6] (ISO/TS 15066). This speed is much lower than the typical value of 1 m/s for a human arm [7], [8], and this results in lower productivity.

To allow the operating speed to be increased, it is important to reduce the inertia of both the robot arm and the driving motor. An effective way to achieve this is the use of a tendon drive, in which motors are placed at the base of the arm and the force is transmitted by wires, as demonstrated in many light-weight robot arms [9], [10], [11]. However, if the inertia of the motor itself is large, simply reducing the arm inertia may not be sufficient.

In a typical combination of a motor and a reduction gear, the motor inertia is multiplied by the square of the reduction ratio. If the reduction ratio is high, the resulting inertia can be very large, which could be even comparable to the arm inertia. To reduce the motor inertia, it might be considered effective to use a direct-drive motor that does not require a reduction gear. However, commercially available direct-drive motors are both heavy and large [12]. To take full advantage of the low-inertia characteristics of direct-drive motors, the motors themselves must be compact and lightweight. As such a compact and lightweight direct-drive motor, this study focuses on an electrostatic film motor [13].

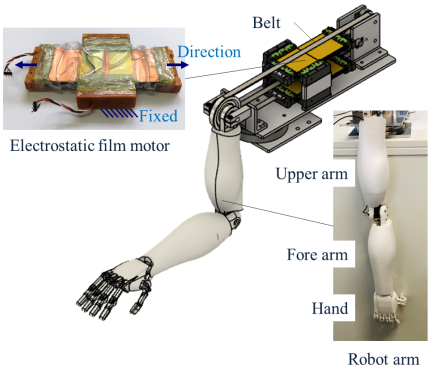


Fig. 1: Prototype robot arm driven by electrostatic film motors.

The electrostatic film motor is a synchronous linear motor driven by a 3-phase ac high voltage. This paper reports on a prototype 2-degree-of-freedom (DOF) robot arm driven by two electrostatic film motors (Fig. 1). The lightness of the motor, combined with the adopted tendon drive mechanism, is expected to lead to a low-inertia arm. In addition, the electrostatic motor can function as a variable toque/force limiter, which is the main focus of this work.

In the electrostatic film motor, the movement speed is determined by the frequency of the applied ac voltage, due to the synchronous characteristics of the motor. The output force is passively determined such that it balances the external force. However, there is a maximum limit to the output force, which is determined by the voltage amplitude, and if the external force exceeds this limit, the motor will "step out" from its synchronous state. In the present study, this step-out behavior is utilized to realize a variable torque/force limiter.

There have been several studies on variable torque limiters, often utilizing a magneto-rheological fluid [14], [15]. However, magneto-rheological devices suffer from sedimentation of magnetic particles, which can deteriorate their performance. An electrostatic film motor can act as a variable torque limiter without the need for magnetic particles. Another conventional problem with using a torque limiter is

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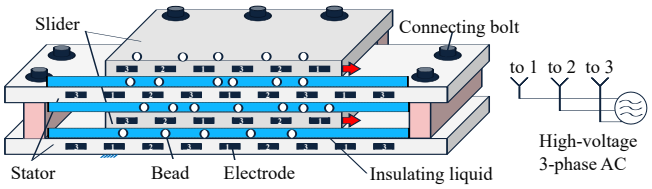


Fig. 2: Cross section of electrostatic film motor.

that once the limiter is activated, it is difficult for the arm to resume normal operation. If, however, the motor itself has a torque-limiting function, resuming normal arm motion can be simply achieved.

This paper demonstrates the advantages of using electrostatic film motors in the developed prototype arm. The prototype arm has a variable torque-limiting function that allows it to perform necessary movements while avoiding excessive contact force, which is not possible when using a high-reduction-ratio gear and an electromagnetic motor. The structure of the electrostatic film motor is described in the next section. Section III describes the design of the prototype arm. The performance of the variable torque limiter is experimentally demonstrated in section IV. Section V provides the conclusions of this study.

II. Electrostatic Film Motor as Torque Limiter

A. Electrostatic film motor

The electrostatic film motor used in this work consists of two sets of films, which are referred to as a stator and a slider (Fig. 2). In the stator and the slider films, fine electrodes are aligned at a regular interval, which is referred to as the electrode pitch p and is 200 μm in this work, to produce three-phase electrode structures. The films are immersed in a dielectric liquid to allow the use of a high electric field without discharge occurring. For motor operation, a 3-phase ac high voltage, with an amplitude of around 1 to 2 $kV_{0\text{-}p}$, is applied to the three-phase electrodes.

The output force capability is proportional to the combined area of the interfaces between the slider and the stator films. Therefore, stacking several films as in Fig. 2 is an effective means of increasing the output force capability. For example, in a previous study, an output force of 300 N was achieved using 50 pairs of films [13].

When a 3-phase ac voltage, $V[\sin(\phi), \sin(\phi - 2\pi/3), \sin(\phi + 2\pi/3)]$, is applied, the motor generates a force expressed as follows [16]:

$$f = \kappa V^2 \sin(\delta) = \kappa V^2 \sin(\theta - 2\phi) \qquad (1)$$

where κ is a force coefficient determined by the film area. In the equation, θ represents the slider position expressed as an electric angle, in which the period of the 3-phase electrodes, $3p$, is normalized to 2π. In other words, $\theta = 2\pi x/3p$ where x is the actual position. The angle $\delta = \theta - 2\phi$ is referred to as the force angle.

B. Synchronous operation and step-out

In a synchronous operation, where the motor is operated using a 3-phase ac voltage, the slider displacement is determined by the phase ϕ of the applied voltage. Figure 3 shows the force generated by the motor based on Eq. 1 to illustrate synchronous operation. During synchronous operations, stable equilibrium is achieved when the force angle δ is such that the output force intersects with the external force with a negative slope. If the external force is constant, the force angle, $\delta = \theta - 2\phi$, is also constant. Therefore, as the phase ϕ changes with time, the slider position θ also changes so as to satisfy $\theta = 2\phi + \delta$.

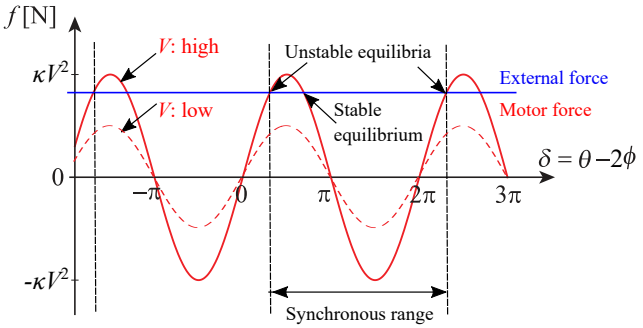


Fig. 3: Force produced by synchronous electrostatic motor (red lines) and applied external force (blue line).

The force angle δ changes according to the external force. If the external force increases, δ will shift such that the motor output force will also increase. However, if the external force exceeds κV^2, equilibrium is lost and the slider will drift to escape from the external force. This step-out effect is observed for all synchronous drives, not just for an electrostatic film motor.

C. Variable torque limiter

During step-out, the slider is moved by the external force until the force acting on the motor becomes smaller than the maximum force. Such behavior is similar to that of a torque limiter, and thus can be used to limit the force. Moreover, since step-out occurs when the external force exceeds the maximum force, κV^2, the limit value can be changed using the voltage amplitude V.

The voltage amplitude V is controlled such that κV^2 is slightly larger than the force required to accomplish a task. This κ value can be modeled based on the capacitances of an electrostatic film motor [16], but to determine the force more accurately, κ is obtained from the actual force measured when a specific voltage is applied. If there is an unexpected disturbance during a task, the motor will automatically step out to escape from the disturbance. In such a way, the motor guarantees safer contact with the environment. In the present study, the motor is used to drive the rotational joints in a 2-DOF robot arm. A limit is set for the maximum torque calculated for each joint.

D. Position feedback control

In a synchronous motor, open-loop position control can be used for the motor trajectory, since the slider position is determined by the phase of the applied voltage. However, if step-out occurs, the slider deviates from the target position, and closed-loop position control is required.

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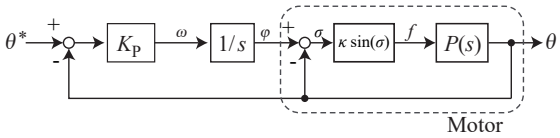


Fig. 4: Position controller for electrostatic film motor.

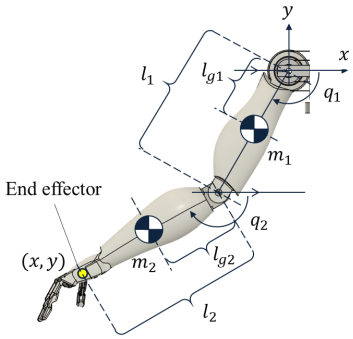


Fig. 5: Coordinates and symbols for robot arm.

This is achieved by simple feedback of the potion error to the frequency $\omega = d\phi/dt$ of the applied ac voltage. Figure 4 shows the control block diagram. The slider position, θ, is measured by an encoder, and its error relative to the reference, θ^*, is provided to a proportional controller with a gain of K_p. The output of the proportional controller is a frequency command, which is integrated to obtain a phase command. The three-phase ac voltage required to achieve the desired phase is then calculated and applied to the motor. From another viewpoint, this controller can be seen as an integral controller for the phase ϕ, so it can be easily understood that it can eliminate position errors.

When an external force is smaller than κV^2, the output of the controller, ω, is almost proportional to the speed of the reference position. The resulting waveform and the motor behavior are similar to those in open-loop operation. When step-out occurs, there is a large position error. In this case, the frequency, ω, increases to speed up the motor so it can catch up with the reference position after the external disturbance disappears. In such a way, the arm can quickly resume its original trajectory.

III. Prototype Robot Arm

A. Arm design

The 2-DOF prototype robot arm was designed to have a length of about 0.5 m, which is similar to that of a human arm. The links were made of ABS plastic fabricated using a 3D printer, and the movable angle range for both joints was 90 degrees. Figure 5 shows the major parameters for the robot arm. The mass of the upper arm (m_1) was 0.165 kg, and that of the forearm including the hand (m_2) was 0.192 kg. The lengths of the links for the upper arm and the forearm (l_1 and l_2) were both 0.2 m. The center of mass (COM) of the upper arm was located at $l_{g1} = 0.091$ m from the shoulder joint. The COM of the forearm was located at $l_{g2} = 0.11$ m from the elbow joint. The moment of inertia around COM of the upper arm (I_1) and the forearm link (I_2) was $5.9\times10^{-4}/1.5\times10^{-3}$ kg·m^2, estimated from the geometry of the CAD model and the measured weight and length.

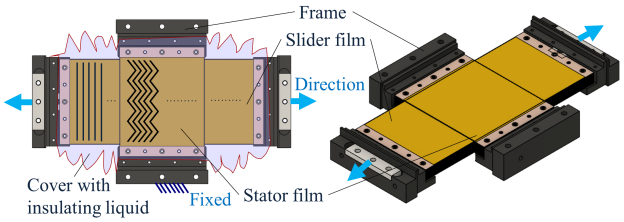


Fig. 6: Structure of 25-layered electrostatic film motor. The shorter films were fixed to the base and thus acted as stators.

B. Multi-layer electrostatic film motors

The motors used for the robot were designed such that the arm can lift a 300 g object from its lowest position to a horizontal position. To fulfill this requirement, two motor modules were developed with identical designs, as shown in Fig. 6. In each module, 25 pairs of stator and slider films were stacked. The slider film (which is the longer one) contained straight electrodes, whereas the stator film contained skewed (zigzag) electrodes for smooth motion characteristics [16]. Both electrodes were aligned with an electrode pitch of 200 μm. A double-sided film design [17] was adopted for the films.

To reduce friction between the films and maintain an appropriate gap between them, 30 μm glass beads were scattered in all the gaps between the films. The films were bound by plastic frames and then encapsulated in a flexible plastic case (bellows) filled with a dielectric liquid. As the dielectric liquid, Fluorinert FC-3283 (3M) was selected for its low viscosity (1.44 mPa·s).

The total size of the motor module was 150 mm×100 mm×13 mm and its weight was 280 g including the insulating liquid. The slider, which is the moving mass, weighed approximately 100 g in total, with each slider film being about 1 g.

Each motor module was driven using a 3-phase voltage generated by a position controller. Both the controller and the signal generator were implemented on a digital signal processing (DSP) board (DSpace, ds1104). The 3-phase signals from the DSP board were amplified 1000 times by using three high-voltage amplifiers (NF Corporation, HV-4321). Thus, a total of six amplifiers were used in total to drive the two modules.

The force coefficient κ for the motor modules was measured using a force sensor connected between the end of the slider frame (Fig. 6) and a fixed wall. The applied 3-phase ac voltage had a frequency of 1 Hz, corresponding to a synchronous movement speed of 1.2 mm/s. The voltage amplitude was varied from 0.5 to 1.6 kV in 0.1 kV increments and the maximum force for each voltage was measured. The results are shown in Fig. 7. The plot is not completely parabolic, as predicted by the force equation, possibly due to changes in the motor parameters during the measurements. Based on the results, a force coefficient of $\kappa = 40$ μN/V^2 was used in subsequent experiments.

When combined with the tendon drive described in the next section, a shoulder torque of 2.2 N·m is expected for an

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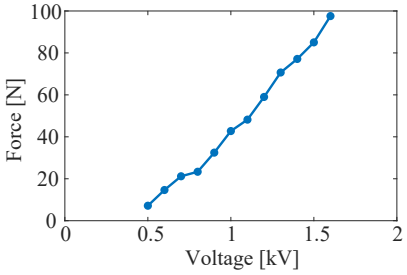


Fig. 7: Dependence of maximum output force on voltage amplitude for electrostatic motor module.

applied voltage of 1.3 kV. Since the shoulder torque required to lift a payload of 300 g is about 1.9 N·m, the motor is strong enough to meet the requirement.

C. Tendon-driven structure

To reduce the arm inertia, a tendon-drive mechanism was adopted. The motors were placed at the base of the arm, and the motor output force was transmitted by tendons, as shown in Fig. 8. To overcome problem of wire stretching associated with typical tendon drives, thin lightweight rigid steel belts (Dymco, D1184) were used as tendons. The belts were 10-mm wide and 50-μm thick, and had a tensile strength of more than 500 N.

A belt was connected to both ends of the slider frame of the motor and formed a closed loop through aluminum pulleys with built-in bearings. For the shoulder joint (joint 1), motor 1 was connected to the upper arm link, through a pulley with a radius of r_1. The pulley was fixed to the upper arm link, and the slider motion of motor 1 was converted to rotary motion of the upper link.

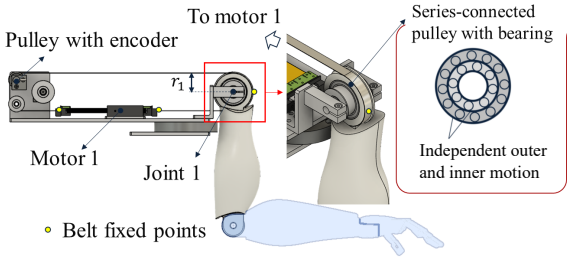
To the elbow joint (joint 2), motor 2 was connected using two steel belts, through two pulleys with a radius of r_2. The slider motion of motor 2 was converted to rotary motion of the forearm link around joint 2. To avoid interference between the two motor motions, the pulleys at the shoulder were arranged co-axially at the base of the upper arm, such that they could rotate independently (see upper right inset in Fig. 8(a)).

The pulley radius was $r_1 = 28$ mm for joint 1, and $r_2 = 15$ mm for joint 2. Within these radii, the equivalent inertia of the motor was calculated to be 7.8×10^{-5} kg·m^2 for joint 1 and 2.3×10^{-5} kg·m^2 for joint 2.

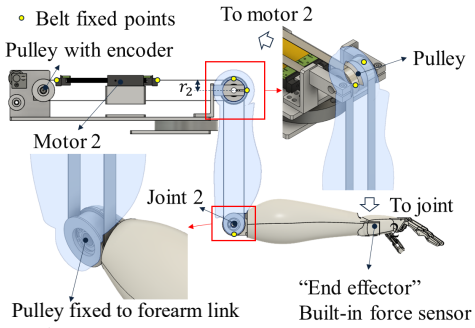
To measure the joint angles (which are equivalent to the slider positions), two encoders (EXTCOM, EV10S) were attached to the pulleys installed in the base. A single-axis force sensor (Minebea, CM085-200N) was installed to measure the contact force on the back of the hand, as shown in Fig. 8(b).

D. Calculation of trajectory and torque limit

When driving the robot arm, a target trajectory was given to the end-effector position (x, y), from which the target joint angles were calculated by inverse kinematics. These angles were then converted to motor displacements, which were used as references for the position feedback controllers that determined the phases of the applied ac voltages.



(a) Structure around joint 1.



(b) Structure around joint 2.

Fig. 8: Structure of robot arm using tendon drive. Two motors at the base drive the arm via belts.

The applied ac voltage amplitudes were set considering the torque limit. First, the torques required to accomplish a task were calculated. Torques $\tau_1(t)$ and $\tau_2(t)$, required to move the arm along the target trajectory, were calculated by inverse dynamics, and the internal losses associated with the motor were then added. These internal losses were assumed to be mainly caused by the viscosity of the liquid. The viscous forces for the two motors were estimated, and were converted into joint torques τ_{d1} and τ_{d2}. The viscous force for the motor was assumed to be proportional to the motor speed and was $dr_1\dot{q}_1$ for motor 1, or $\tau_{\mathrm{d1}} = dr_1^2\dot{q}_1$ if expressed as a joint torque. The viscous coefficient d was estimated as

$$d = \frac{\eta(2n-1)S}{y_0} = 59\ [\mathrm{N \cdot s/m}] \tag{2}$$

using a liquid viscosity of $\eta = 14.4 \times 10^{-3}$ N·s/m^2, an effective area of $S = 25 \times 10^{-4}$ m^2, an estimated film gap of $y_0 = 30 \times 10^{-6}$ m, and $n = 25$ layers. Finally, torque margins τ_{m1} and τ_{m2} were added. The resulting torque limit for joint 1 was $\tau_{\mathrm{tot1}} = \tau_1 + \tau_{\mathrm{d1}} + \tau_{\mathrm{m1}}$ and τ_{tot2} for joint 2.

The calculated torque limit for motor 1 was then converted back to motor force using

$$F_1(t) = \frac{\tau_{\mathrm{tot1}}(t)}{r_1} = \frac{\tau_1(t) + \tau_{\mathrm{m1}} + dr_1^2\dot{q}_1(t)}{r_1}. \tag{3}$$

Based on the force limit, the voltage amplitude was set as

$$V_1(t) = \sqrt{\frac{F_1(t)}{\kappa_1}} = \sqrt{\frac{\tau_1(t) + \tau_{\mathrm{m1}} + dr_1^2\dot{q}_1(t)}{r_1\kappa_1}}, \tag{4}$$

where κ_1 is the force coefficient for motor 1. The same calculation was performed for motor 2.

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IV. EXPERIMENTS

A. Position trajectory tracking

Position-tracking experiments were conducted using the prototype robot arm. The robot arm was controlled such that the end effector followed a circular trajectory with a radius of 0.05 m, centered at $(-0.25, -0.2)$, travelling counterclockwise at a constant angular velocity of π rad/s. The position of the end effector was estimated from the two encoders. The position control gains were set to 10 s^{-1} for both motors. The torque margins for the two joints were set as $\tau_{m1} = \tau_{m2} = 0.5$ N·m.

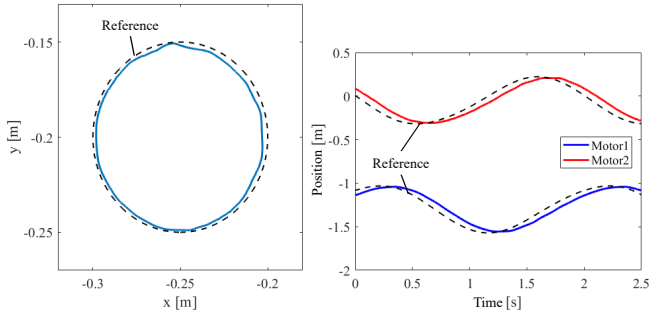
The results are shown in Fig. 9. The end effector traced the target trajectory with an error of about 2 mm, which is 2% of the trajectory diameter. It was found that the motion of each motor was delayed relative to the reference by about 0.1 s, which was the cause of the trajectory error. This delay was probably due to the simple proportional feedback controller adopted in this work and would be improved by introducing additional feedback actions and fine-tuning the gains.

B. Impact resistance due to torque limiter

Since the torque limiter can avoid excessive contact force in unexpected collisions, the developed robot arm is impact-resistant. To demonstrate this, the end effector was struck hard using a hammer, as seen in Fig. 10(a).

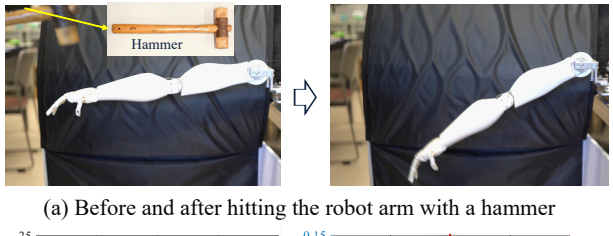
In this experiment, the robot arm was commanded to maintain a horizontal posture. Feedback gains of 10 s^{-1} were used for the two motors, together with torque margins of $\tau_{m1} = \tau_{m2} = 0.5$ N·m. During arm motion, the hammer struck the back of the hand where the force was being monitored. The collision force measured by the force sensor is shown in Fig. 10(b). The hammer tip weighed 0.47 kg and the velocity before the collision was estimated to be about 10 m/s from a video analysis. Therefore, if the arm had no torque limiter and was not back-drivable, a maximum force of about 470 N might have been exerted, assuming a contact duration of 20 ms. However, the measured peak contact force was only about 23 N, which means that it was significantly suppressed by the torque limiter.

The experimental results also show the capability of the arm to resume its original position. When the hammer struck, the end effector moved +5 cm in the x direction and −12 cm in the y direction as seen in Fig. 10(c). However, about 0.1 s after the hammer striking, the feedback controllers began to restore the arm to its original positions, which completed in about 0.5 s. This result shows that the robot arm and the motors maintained their functionality even after being struck hard by the hammer.

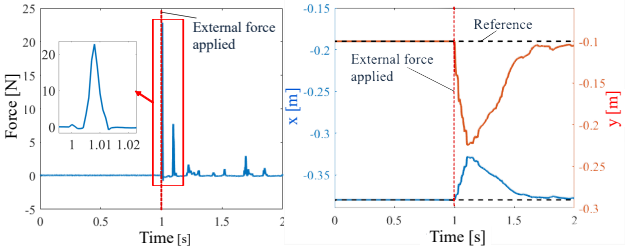


(a) Position (x, y) of end effector (b) Time series motor position

Fig. 9: Position (x, y) of end effector (estimated from encoder outputs) and corresponding motor positions.



(a) Before and after hitting the robot arm with a hammer



(b) Contact force. (c) End effector position x, y.

Fig. 10: (a) Photographs before and after end effector was struck by hammer, plots of (b) contact force and (c) end-effector position x, y (estimated from encoder values).

C. Dynamic torque limiter

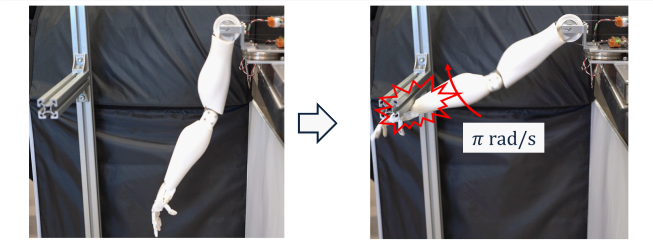
In the next experiment, the torque limiter was tested during a high-speed motion. The robot arm was commanded to follow a circular trajectory with a radius of 0.395 m, centered on the shoulder axis (joint 1). The commanded angular velocity was π rad/s, which is equivalent to a circumferential velocity of 1.2 m/s. An aluminum pillar (3 cm × 3 cm) was placed in the middle of the circular trajectory, such that the moving hand hit the pillar, as seen in Fig. 11(a). The position controller gain was 10 s^{-1} for both motors. Experiments were conducted with three different torque margins of $\tau_{m1} = \tau_{m2} = 0.1, 0.3, 0.5$ N·m.

The contact force and the end effector position are shown in Fig. 11(b) and (c), respectively. For the lowest torque margin (0.1 N·m), the robot arm could not swing due to insufficient output force. This is likely to be because of imperfect modeling of the motor output force, as well as imperfect estimation of the force loss in the setup; the torque margin was not sufficient to compensate for these imperfections.

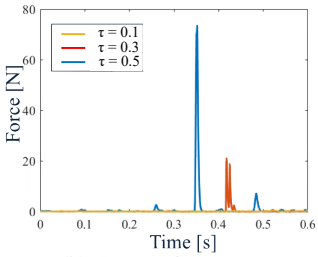
When the torque margin was larger (0.3 and 0.5 N·m), the arm was successfully lifted and hit the aluminum pillar. Although some small spikes are observed in the force curves plotted in Fig. 11(b), the largest spike represents the collision, and the others are probably noise. The contact timing differs between the two different torque margin settings, even though the reference velocity was the same.

The peak contact force was 21 N for $\tau_{m1} = \tau_{m2} = 0.3$ N·m and 74 N for $\tau_{m1} = \tau_{m2} = 0.5$ N·m. This indicates that the contact force correlated with the torque margin. It has been reported [18] that a contact force of up to 50 N

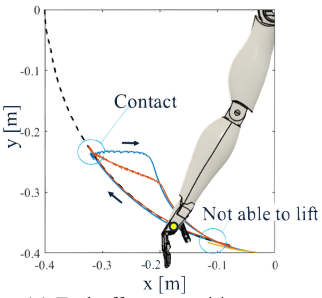
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(a) High-speed contact with the robot hand.



(b) Contact force.



(c) End effector position x, y.

Fig. 11: Robot arm collision experiment: (a) photographs of contact with pillar, (b) measured contact force, and (c) effector position estimated from encoder outputs.

is tolerable for humans, so the contact force of 21 N in the present study (for $\tau_{m1} = \tau_{m2} = 0.3$) can be considered safe. It would be difficult to suppress the contact forces in this way using a combination of a high-reduction-ratio gearbox and an electromagnetic motor, as used in a typical robot arm [19]. Although the results are still preliminary, they indicate that by setting an appropriate torque margin, the contact force can be kept at a safe level for humans even under high-speed operation.

V. CONCLUSIONS

To realize a robot arm that can achieve both high-speed motion and safe contact, this study proposed the use of electrostatic film motors, which are lightweight direct-drive motors. In the proposed configuration, a tendon-driven mechanism reduces the arm inertia and the direct-drive motors reduce the motor inertia, thereby reducing the overall inertia of the robot arm. In addition, by utilizing the step-out behavior of synchronous motors, a variable torque limiter was proposed whose limit value can be controlled by the applied voltage amplitude.

To demonstrate the proposed torque limiting function, a prototype 2-DOF robot arm weighted 350 g and was 0.5 m in length was developed using two electrostatic film motors. Thanks to the toque limiter, the developed arm proved to be impact-resistant. Even when it was struck by a hammer, the arm recovered from the collision due to torque limiter, thus reducing the contact force. In a second experiment, a robot arm was moving at a speed higher than 1 m/s hit an obstacle. The resulting contact force was 21 N, which is less than the maximum tolerable contact force of 50 N for humans. These results verified the effectiveness of the proposed setup, especially the variable torque limiter, for realizing safe robot arm operations.

The prototype robot arm described in this paper was developed for verification of the principles involved, and there are still many issues to be solved. However, the results clearly show the advantages of utilizing lightweight synchronous motors in robot arms, and indicate another approach to realizing safe robots in the future. It should be noted that the proposed concept is not limited to electrostatic film motors, but would apply to any lightweight synchronous motor. Developing such a new motor is a subject for future work.

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